

MODULE NO: 3

MATHEMATICAL LOGICS

* Definition:

* Proposition:

- A proposition is a declarative statement which is either true or false but not both.
- Propositions are usually denoted by small letters such as p, q, r, s , etc.
- If the proposition is true then the truth value of the proposition is 1 and if the proposition is false then the truth value is 0.

Example:

- $\sqrt{5}$ is an integer $\rightarrow 0$
- $\frac{-1}{3}$ is a real number $\rightarrow 1$
- 0 is a natural number $\rightarrow 0$
- $-2 \cdot 3$ is an integer $\rightarrow 0$

* Logical connectivities:

- The words like NOT, AND, OR, IF, IFF, etc are called logical connectives.
- Any statement containing these logical connectivities are called Compound Propositions.
- The propositions which are free from logical connectivities are called Simple Propositions.

i. Negation - NOT ($\sim$)

Let p be any proposition then the negation of p is given by $\sim p$. If the truth value of p is true then the truth value of $\sim p$ is false. Similarly if the truth value of p is false then the truth value of $\sim p$ is true.

iii. Disjunction:- OR ($\vee$)

Let p and q be any two propositions then the disjunction of p and q is given by $p \vee q$.
The truth value of $p \vee q$ is false only when both p as well as q are false otherwise it is true.

Truth table:

Example

p	q	$p \vee q$	
1	1	1	-3 is an int or -3 is real
1	0	1	-3 is an int or -3 is not real
0	1	1	-3 is not an int or -3 is real
0	0	0	-3 is not an int or -3 is not real

where

 $p = -3$ is an integer $q = -3$ is realiv. Conditional - if-then ($\rightarrow$)

Let p and q be any two propositions then the conditional of p and q is given by $p \rightarrow q$.
The truth value of $p \rightarrow q$ is false only when p is true and q is false otherwise it is true.

Truth table:

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

Example:

If it is raining then we will not play.
If it is raining then we will play.

proposition-1

a Natural number
is not a Natural number

Identify the no. of terms in the table.

 $n = \text{no. of propositions}$

2

function:- AND ($\wedge$)

p and q be any two propositions then the conjunction of p and q is given by $p \wedge q$.
The truth value of $p \wedge q$ is true only when both p as well as q are true otherwise it is false.

Truth table:

Example

p	q	$p \wedge q$	
1	1	1	-3 is an int & -3 is Real
1	0	0	-3 is an int & -3 is not Real
0	1	0	-3 is not an int & -3 is Real
0	0	0	-3 is not an int & -3 is not Real

where

 $p = -3$ is an integer $q = -3$ is Real

If it is not raining then we will play

If it is not raining then we will play

v. Biconditional - Iff ($\leftrightarrow$)

Let p and q be any two propositions then the biconditional of p and q is given by $p \leftrightarrow q$.

The truth value of $p \leftrightarrow q$ is true only when both p and q have same truth value i.e., both 1 and both 0 otherwise it is false.

Truth table:

	p	q	$p \leftrightarrow q$
i.	1	1	1
ii.	1	0	0
iii.	0	1	0
iv.	0	0	1

Example:

- i. $x + 5 = 10$ if and only if $x = 5$
- ii. $x + 5 = 10$ if and only if $x = 4$
- iii. $x + 5 = 10$ if and only if $x = 5$
- iv. $x + 5 = 10$ if and only if $x = 4$

NOTE:

- * A Compound proposition which is true for all possible truth values is known as Tautology (T_0).
- * A Compound proposition which is false for all possible truth values is known as Contradiction (F_0).
- * A Compound proposition which is either true or false or both is known as Contingency

v.

p	q	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
1	1	0	0	0
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

vi.

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
1	1	1	1
1	0	1	1
0	1	1	1
0	0	0	1

It is Tautology.

vii.

p	q	$\sim p$	$(\sim p \wedge q)$	$p \wedge (\sim p \wedge q)$
1	1	0	0	0
1	0	0	0	0
0	1	1	1	0
0	0	1	0	0

It is Contradiction.

viii.

X

p	q	r	$\sim q$	$(\sim q \rightarrow p)$	$\sim p$	$(\sim q \rightarrow p) \leftrightarrow \sim p$
1	1	1	0	1	0	0
1	1	0	0	1	0	0
1	0	1	1	1	0	0
1	0	0	1	1	0	0
0	1	1	0	1	1	0
0	1	0	0	0	1	1
0	0	1	1	1	1	0
0	0	0	1	1	1	0

vii.

p	q	$\sim q$	$(\sim q \rightarrow p)$	$\sim p$	$(\sim q \rightarrow p) \leftrightarrow \sim p$
1	1	0	1	0	0
1	0	1	1	0	0
0	1	0	1	1	0
0	0	1	0	1	1

(5m) ix.

Bin property
L. Exponent
Theorem

p	q	$p \vee q$	$(\sim p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	1	1	0	0
0	0	0	1	1	1	1

$\therefore A \leftrightarrow B //$

x.

p	q	r	$(p \wedge q)$	$(\sim p \wedge \sim q)$	$(p \wedge q) \vee (\sim p \wedge \sim q)$	$[(p \wedge q) \vee (\sim p \wedge \sim q)] \leftrightarrow p$
1	1	1	1	0	1	1
1	1	0	1	0	1	1
1	0	1	0	0	0	0
1	0	0	0	0	0	0
0	1	1	0	1	1	0
0	1	0	0	1	1	0
0	0	1	0	1	1	0
0	0	0	0	1	1	0

xi. PT $[(\neg q) \wedge (p \rightarrow q)] \rightarrow \sim p$ is T_0

p	q	($\neg q$)	$p \rightarrow q$	$[(\neg q) \wedge (p \rightarrow q)]$	$\sim p$	$[(\neg q) \wedge (p \rightarrow q)] \rightarrow \sim p$
1	1	0	1	0	0	1
1	0	1	0	0	0	1
0	1	0	1	0	1	1
0	0	1	1	1	1	1

$\therefore H$ is $T_0 //$

xii. PT $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is T_0

p	q	r	$(p \rightarrow q)$	$(q \rightarrow r)$	$[(p \rightarrow q) \wedge (q \rightarrow r)]$	$(p \rightarrow r)$
1	1	0	1	0	0	0
1	0	1	0	1	0	1
1	0	0	0	1	0	0
0	1	1	1	1	1	1
0	1	0	1	0	0	1
0	0	1	1	1	1	1
0	0	0	1	1	1	1

$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

$\therefore H$ is $T_0 //$

xiii. PT $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$

p	q	r	$(p \vee q)$	$[(p \vee q) \rightarrow r]$	$(p \rightarrow r)$	$(q \rightarrow r)$	$[(p \rightarrow r) \wedge (q \rightarrow r)]$
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	0	1	1	1	1	1	1
1	0	0	1	0	0	1	0
0	1	1	1	1	1	1	1
0	1	0	1	0	1	0	0
0	0	1	0	1	1	1	1
0	0	0	0	1	1	1	1

$[(p \vee q) \rightarrow r] \Leftrightarrow B$
 $[(p \rightarrow r) \wedge (q \rightarrow r)] \quad A \Leftrightarrow$

1
0
1
0
1
0
1
0

$\therefore A \Leftrightarrow B //$

xiv. PT $[(p \vee q) \rightarrow r] \Leftrightarrow [\sim r \rightarrow \sim(p \vee q)]$ is T_0

p	q	r	$(p \vee q)$	$[(p \vee q) \rightarrow r]$	$\sim r$	$\sim(p \vee q)$	$[\sim r \rightarrow \sim(p \vee q)]$
1	1	1	1	1	0	0	1
1	1	0	1	0	1	0	0
1	0	1	1	1	0	0	1
1	0	0	1	0	1	0	0
0	1	1	1	1	0	0	1
0	1	0	1	0	1	0	0
0	0	1	0	1	0	1	1
0	0	0	0	1	1	1	1

* Converse, Inverse and Contrapositive

Let $p \rightarrow q$ be a conditional statement then

- i) $q \rightarrow p$ is called Converse
- ii) $\sim p \rightarrow \sim q$ is called Inverse
- iii) $\sim q \rightarrow \sim p$ is called Contrapositive

Q Write Converse, Inverse, Contrapositive of the following Conditional

- 1) p : lines are parallel
 q : they intersect each other

Sol: $\therefore p \rightarrow q$
If lines are parallel then they intersect each other.

• $q \rightarrow p$
If they intersect each other then the lines are parallel.

• $\sim p \rightarrow \sim q$
If lines are not parallel then they do not intersect each other.

• $\sim q \rightarrow \sim p$
If they does not intersect each other then the lines are not parallel.

- 2) p : Quadrilateral is parallelogram
 q : Diagonals of quadrilateral bisect each other.

• $p \rightarrow q$
If quadrilateral is parallelogram then diagonals of quadrilateral bisect each other.

$$[(p \vee q) \rightarrow r] \leftrightarrow [\neg r \rightarrow \neg(p \vee q)] \text{ is } T_0$$

↓
↓
↓
↓
↓
↓
↓

$A \leftrightarrow B$ is T_0 . It is T_0 //

xiv. Let p and q

xv. Let p and q be primitive statements for which the conditional $p \rightarrow q$ is false. Determine the truth values of the following compound proposition.

1. $p \wedge q$
2. $\sim p \vee q$
3. $q \rightarrow p$
4. $\sim q \rightarrow \sim p$

Sol: Given: $p \rightarrow q = 0$
 $\therefore p = 1 \text{ \& } q = 0$

1. $p \wedge q \Rightarrow 1 \wedge 0 = 0$
2. $\sim p \vee q \Rightarrow 0 \vee 0 = 0$
3. $q \rightarrow p \Rightarrow 0 \rightarrow 1 = 1$
4. $\sim q \rightarrow \sim p \Rightarrow 1 \rightarrow 0 = 0$

xvi. If p is true and q is false. Find the truth values of the following

$$1. (p \wedge q) \vee \sim(q \rightarrow p)$$

Sol: Given: $p = 1 \text{ \& } q = 0$

$$(1 \wedge 0) \vee \sim(0 \rightarrow 1)$$

$$0 \vee \sim 1$$

$$0 \vee 0$$

$$0 //$$

$$2. (p \rightarrow q) \vee \sim(p \leftrightarrow \sim q)$$

Given: $p = 1 \text{ \& } q = 0$

$$(1 \rightarrow 0) \vee \sim(1 \leftrightarrow \sim 0)$$

$$0 \vee \sim(1 \leftrightarrow 1)$$

$$0 \vee \sim 1$$

$$0 \vee 0$$

$$0 //$$

xvii. Indicate how many rows and columns are needed in the truth table for the compound proposition $(p \vee \sim q) \leftrightarrow [(\sim r \wedge s) \rightarrow t]$. Find the truth value of proposition if p and r are true and q, s, t are false.

Sol: The no. of rows $= 2^n = 2^5 = 32$.

Given $p = 1, r = 1, q = 0, s = 0, t = 0$.

$$p[(1 \vee \sim 0) \leftrightarrow [(\sim 1 \wedge 0) \rightarrow 0]]$$

$$(1 \vee 1) \leftrightarrow [(0 \wedge 0) \rightarrow 0]$$

$$1 \leftrightarrow [0 \rightarrow 0]$$

$$1 \leftrightarrow 1$$

$$1$$

Open Statement:

A statement which becomes a proposition when the value of the variables involve are known is known as

Open Statement.

$P(x): x+3=5$ } Open Statement

$x=1: P(1): 1+3=5 \Rightarrow F=0$ } Proposition

$x=2: P(2): 2+3=5 \Rightarrow T=1$ }

Quantified Statement:

A proposition involving the idea of a quality is known as quantified statement.

All the sentences consisting of words like for all, for every, for some etc.

There are 2 types of Quantifiers

Universal Quantifier

If the phrases like for all, for every, for each are involved in a sentence then it is known as Universal Quantifier. It is denoted by $\forall$.

Existential Quantifier:

If the phrases like for some, there exist, atleast are involved in a sentence then it is known as Existential Quantifier. It is denoted by $\exists$.

Problem:

Let the Universal set be the set of all integers. Consider the following Open Statement

$x): x \leq 3$

$x): x+1$ is odd

$x): x \leq 0$

the truth value of the following.

(i) $P(2)$

Given - $P(x): x \leq 3$

$P(2): 2 \leq 3 = T$

$P(2): 1$

(ii) $P(3) \vee \sim r(3)$

Given: $P(x): x \leq 3$

$P(3): 3 \leq 3 = T$

$\therefore P(3) = 1$

$r(x): x \leq 0$

$r(3): 3 \leq 0 = F$

$\therefore r(3): 0$

$\therefore P(3) \vee \sim r(3)$

$1 \vee 0$

$1 \vee 1$

$1 //$

(iii) $q(-9)$

Given: $q(x): x+1$ is odd

$q(-9): -9+1$ is odd

$q(-9): -8 = F$

$\therefore q(-9) = 0$

(iv) $\sim p(4) \wedge \sim q(5)$

Given: $p(x): x \leq 3$

$p(4): 4 \leq 3 = F$

$p(4) = 0$

$\sim p(4) =$

$\Rightarrow \sim p(4) \wedge \sim q(5)$

$1 \wedge 1$

$1 //$

$q(x): x+1$ is odd

$q(5): 5+1$ is odd

$q(5): 6 = F$

$\therefore q(5) = 0$

Q. Express the following symbolic statements in words and indicate the truth value.

I. $\forall x$ for the set of all integers $P(x): x > 0$
 $q(x): x$ is even
 $r(x): x$ is a perfect square
 $s(x): x$ is divisible by 3
 $t(x): x$ is divisible by 7

Sol: For all integers, If x is a perfect square then $x > 0$.

$$x = 0, r(x) \rightarrow P(x)$$

$$r(0) \rightarrow P(0) \rightarrow 0 > 0$$

$$0 \rightarrow 0$$

0 //

2. $\exists x, [s(x) \wedge \sim q(x)]$ where $x = 9$

Sol: for some ^{integer} x is divisible by 3 and x is not even.

$$x = 9, [s(x) \wedge \sim q(x)]$$

$$s(9) \wedge \sim q(9)$$

$$1 \wedge \sim 0$$

$$1 \wedge 1$$

1 //

3. $\forall x, [\sim r(x)]$ $x = 4$

Sol: for every ^{integer} x , $[\sim r(x)]$ x is not a perfect square.

$$x = 4, [\sim r(x)]$$

$$[\sim r(4)]$$

$$\sim 1$$

0 //

Order = No. of Row $\times$ No. of Column

$$= 2 \times 2$$

$$= 4 //$$

① $A = \begin{bmatrix} 1 & -3 \\ 6 & 7 \\ 4 & 0 \end{bmatrix}_{3 \times 2}$

② $B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{2 \times 4}$

③ $C = [1 \ -3 \ 4 \ 7 \ 8]_{1 \times 5}$

④ $D = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}_{3 \times 1}$

* Addition and Subtraction of Matrix

Rules

- ① Order must be same
- ② Add / Sub corresponding elements (same position)

Q.1 Find $A+B$, $A-B$

$A = \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 6 \\ 0 & 4 \end{bmatrix}$

$A+B = \begin{bmatrix} 1+8 & 3+6 \\ 7+0 & 9+4 \end{bmatrix} \Rightarrow \begin{bmatrix} 9 & 9 \\ 7 & 13 \end{bmatrix} //$

$A-B = \begin{bmatrix} 1-8 & 3-6 \\ 7-0 & 9-4 \end{bmatrix} \Rightarrow \begin{bmatrix} -7 & -3 \\ 7 & 5 \end{bmatrix} //$

$$A+B = \begin{bmatrix} 1+7 & -7+9 \\ 6+0 & -11+(-49) \\ -11+67 & 4+119 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 6 & -60 \\ 56 & 123 \end{bmatrix} //$$

$$A-B = \begin{bmatrix} 1-7 & -7-9 \\ 6-0 & -11-(-49) \\ -11-67 & 4-119 \end{bmatrix} = \begin{bmatrix} -6 & -16 \\ 6 & 38 \\ -78 & -115 \end{bmatrix} //$$

$$A = \begin{bmatrix} 1 & -8 & -91 \\ 0 & -46 & 78 \\ 66 & 44 & 128 \end{bmatrix} \quad B = \begin{bmatrix} -89 & -97 & 41 \\ 0 & -41 & 66 \\ 78 & 102 & -91 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 1+(-89) & -8+(-97) & -91+41 \\ 0+0 & -46+(-41) & 78+66 \\ 66+78 & 44+102 & 128+(-91) \end{bmatrix} = \begin{bmatrix} -88 & -105 & -50 \\ 0 & -87 & 144 \\ 144 & 146 & 37 \end{bmatrix}$$

$$A-B = \begin{bmatrix} 1-(-89) & -8-(-97) & -91-41 \\ 0-0 & -46-(-41) & 78-66 \\ 66-78 & 44-102 & 128-(-91) \end{bmatrix} = \begin{bmatrix} 90 & 89 & -132 \\ 0 & -5 & 12 \\ -12 & -58 & 219 \end{bmatrix}$$

In general $m \times n$ matrix is written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & \\ a_{31} & a_{32} & a_{33} & \dots & \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

The elements operating in the i^{th} row and j^{th} column which is represented by a_{ij} where $i = 1, 2, 3, \dots, m$

$j = 1, 2, 3, \dots, n$

Example

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Order is 3×3 .

Note: Order of matrix equals to no. of rows $\times$ no. of columns.

matrix or row vector.

In general, $A = [a_{11}, a_{12}, a_{13}, \dots, a_{1n}]$

Example: $[1 \ 2 \ 3 \ 4 \ 5] = A //$

* Column matrix:

A matrix having only one column i.e., a matrix of order $(m \times 1)$ is called column matrix.

In general, $A = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{m1} \end{bmatrix}$

Example: $A = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 6 \end{bmatrix} //$

* Zero or Null matrix:

A matrix in which all the elements are zero is called zero or null matrix. The null matrix of order $m \times n$ is denoted by $O_{m \times n}$

Example: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} //$

* Square matrix:

A matrix in which no. of rows is equal to no. of columns i.e., a square matrix with n -rows

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

Example: $\begin{bmatrix} 2 & 3 \\ 4 & 8 \end{bmatrix} (2 \times 2)$

Note:

i. The elements $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ of a square matrix are called its diagonal elements and the diagonal along which the elements lie is called principal diagonal elements.

ii. The sum of the elements of principal diagonal element is called trace of a matrix.

i.e., In general, trace of a square matrix of order n is $a_{11} + a_{22} + a_{33} + \dots + a_{nn}$

$$= \sum_{i=1}^n a_{ii}$$

Example $A = \begin{bmatrix} 3 & 4 & 3 \\ 2 & 0 & 1 \\ 1 & 1 & 3 \end{bmatrix}$

$$\text{Trace} = 3 + 0 + 3 = 6 //$$

* Rectangular matrix:

A matrix having unequal no. of rows and columns

Example: $A = \begin{bmatrix} 3 & 1 & 3 \\ 2 & 1 & 8 \end{bmatrix} (2 \times 3)$

elements are zero

In general, $A = [a_{ij}]$ is a diagonal matrix if $a_{ij} = 0$ for $i \neq j$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Scalar matrix:

A diagonal in which all the diagonal elements are equal to scalar matrix.

In general, $A = a_{ij}$ is scalar matrix if $a_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ k & \text{if } i = j \end{cases}$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Unit/Identity matrix:

Scalar matrix in which the principal diagonal element is '1' is called unit matrix.

i.e., $A = [a_{ij}]_{n \times n}$ is a unit matrix if $a_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

Identity matrix of order 'n' is denoted as

$$I_n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Upper triangular matrix:

A square matrix in which all the elements below the diagonal are zero is called UTMT.

$A = [a_{ij}]_{n \times n}$ is an upper triangle matrix if $a_{ij} = 0$ for $i > j$ which is denoted by U .

$$U = \begin{bmatrix} 2 & 3 \\ 4 & 8 \\ 0 & 7 \end{bmatrix}$$

Sol. $5A = 5 \begin{bmatrix} -15 & 6 \\ 71 & -38 \end{bmatrix} = \begin{bmatrix} -75 & 30 \\ 355 & -190 \end{bmatrix};$

$6B = 6 \begin{bmatrix} -32 & 48 \\ 710 & -59 \end{bmatrix} = \begin{bmatrix} -192 & 288 \\ 4260 & 354 \end{bmatrix};$

$\Rightarrow (5A - 6B)$
 $\begin{bmatrix} -75 & 30 \\ 355 & -190 \end{bmatrix} - \begin{bmatrix} -192 & 288 \\ 4260 & 354 \end{bmatrix} = \begin{bmatrix} -75 - (-192) & 30 - 288 \\ 355 - 4260 & -190 - 354 \end{bmatrix}$

$= \begin{bmatrix} 117 & -258 \\ -3905 & -544 \end{bmatrix};$

$\Rightarrow (5A - 6B) + 4x = 0$

$\begin{bmatrix} 117 & -258 \\ -3905 & -544 \end{bmatrix} + 4x = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$4x = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 117 & -258 \\ -3905 & -544 \end{bmatrix}$

$4x = \begin{bmatrix} -117 & 258 \\ 3905 & 544 \end{bmatrix}$

$x = \frac{1}{4} \begin{bmatrix} -117 & 258 \\ 3905 & 544 \end{bmatrix}$

$x = \begin{bmatrix} -29.25 & 64.5 \\ 976.25 & 136 \end{bmatrix}$

Q1. Find x and y if $\begin{bmatrix} 1 & -3 \\ x & 4 \end{bmatrix} = \begin{bmatrix} 1 & y \\ 9 & 4 \end{bmatrix}$

Sol: $y = -3, x = 9 //$

Q2. Find a, b, c, d if $\begin{bmatrix} a-b & 2a+c \\ 2a-b & c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$

$a-b = -1 \rightarrow \textcircled{1}$

$2a+c = 5 \rightarrow \textcircled{2}$

$2a-b = 0 \rightarrow \textcircled{3}$

$c+d = 13 \rightarrow \textcircled{4}$

$2 \times \textcircled{1} \times \textcircled{3}$

$2a - 2b = -2$

$2a - 2b = -2$

$2b = 2$

$b = 1 //$

Subs b in $\textcircled{1}$

$a - 1 = -1$

$\Rightarrow a = -1 + 1 \Rightarrow a = 0 //$

~~$2 \times \textcircled{2} \times \textcircled{3}$~~ Now, $2a+c = 5$

~~$2a+c = 5$~~ $2 \times 0 + c = 5$

~~$a, b \neq 0$~~ $c = 5 - 2 = 3 //$

$c+d = 13$

$3+d = 13$

$\Rightarrow d = 13 - 3$

$= 10 //$

$$\textcircled{2} \quad A+B+C = \begin{bmatrix} 1+4+2 & 5+(-2)+3 & 6+3+1 \\ 7+0+1 & 8+(-1)+2 & 9+2+3 \\ 10+(-3)+1 & 11+4+1 & 12+5+1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 6 & 10 \\ 8 & 9 & 14 \\ 8 & 16 & 18 \end{bmatrix} //$$

$$\textcircled{3} \quad A-B+C = \begin{bmatrix} 1-4+2 & 5-(-2)+3 & 6-3+1 \\ 7-0+1 & 8-(-1)+2 & 9-2+3 \\ 10-(-3)+1 & 11-4+1 & 12-5+1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 10 & 4 \\ 8 & 11 & 10 \\ 14 & 8 & 8 \end{bmatrix} //$$

$$\Rightarrow \begin{bmatrix} 18+2 & 2+10 \\ 8+14 & 6+24 \end{bmatrix} + 2x = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 20 & 12 \\ 22 & 30 \end{bmatrix} + 2x = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow 2x = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 20 & 12 \\ 22 & 30 \end{bmatrix}$$

$$x = \frac{1}{2} \begin{bmatrix} -20 & -12 \\ -22 & -30 \end{bmatrix}$$

$$x = \begin{bmatrix} -10 & -6 \\ -11 & -15 \end{bmatrix} //$$

$$q. \text{ If } A = \begin{bmatrix} -15 & 6 \\ 71 & -38 \end{bmatrix} \quad B = \begin{bmatrix} -32 & 48 \\ 710 & -59 \end{bmatrix}$$

$$(5A - 6B) + 4x = 0$$

of columns in the first matrix is equal to the number of rows in the second matrix

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$

$$A_{2 \times 2} B_{2 \times 1} = \checkmark$$

$$B_{2 \times 1} A_{2 \times 2} = \times$$

Consider, $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2 \times 2}$ $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}_{2 \times 2}$

$$AB = \begin{bmatrix} \text{IRXIC} & \text{IRXIC} \\ \text{IRXIC} & \text{IRXIC} \end{bmatrix}$$

$$= \begin{bmatrix} (a_{11} \times b_{11}) + (a_{12} \times b_{21}) & (a_{11} \times b_{12}) + (a_{12} \times b_{22}) \\ (a_{21} \times b_{11}) + (a_{22} \times b_{21}) & (a_{21} \times b_{12}) + (a_{22} \times b_{22}) \end{bmatrix}$$

Row

- 1) Change of column = space
- 2) Change of row = row
- 3) Change of element = +

Find AB

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A_{2 \times 2} B_{2 \times 2} = \begin{bmatrix} (1 \times 5) + (2 \times 7) & (1 \times 6) + (2 \times 8) \\ (3 \times 5) + (4 \times 7) & (3 \times 6) + (4 \times 8) \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} //$$

$$BA = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} (5 \times 1) + (6 \times 3) & (5 \times 2) + (6 \times 4) \\ (7 \times 1) + (8 \times 3) & (7 \times 2) + (8 \times 4) \end{bmatrix}$$

$$BA = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix} //$$

2) $A = \begin{bmatrix} -9 & 6 \\ -11 & -2 \end{bmatrix}$ $B = \begin{bmatrix} 0 & -39 \\ 48 & -53 \end{bmatrix}$

$$AB = \begin{bmatrix} (-9 \times 0) + (6 \times 48) & (-9 \times -39) + (6 \times -53) \\ (-11 \times 0) + (-2 \times 48) & (-11 \times -39) + (-2 \times -53) \end{bmatrix}$$

$$= \begin{bmatrix} 0 + 288 & 351 + (-318) \\ 0 + (-96) & 429 + 106 \end{bmatrix}$$

$$AB = \begin{bmatrix} 288 & 33 \\ -96 & 535 \end{bmatrix} //$$

$$BA = \begin{bmatrix} 0 & -39 \\ 48 & -53 \end{bmatrix} \begin{bmatrix} -9 & 6 \\ -11 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} (0 \times -9) + (-39 \times -11) & (0 \times 6) + (-39 \times -2) \\ (48 \times -9) + (-53 \times -11) & (48 \times 6) + (-53 \times -2) \end{bmatrix}$$

$$= \begin{bmatrix} 0 + 429 & 0 + 78 \\ -432 + 583 & 288 + 106 \end{bmatrix}$$

$$BA = \begin{bmatrix} 429 & 78 \\ 151 & 394 \end{bmatrix} //$$

$$= \begin{bmatrix} (1 \times 1) + (2 \times -8) & (1 \times 0) + (2 \times 16) & (1 \times 2) + (2 \times 11) \\ (-3 \times 1) + (4 \times -8) & (-3 \times 0) + (4 \times 16) & (-3 \times 2) + (4 \times 11) \end{bmatrix}$$

$$= \begin{bmatrix} 1-16 & 0+32 & 2+22 \\ -3-32 & 0+64 & -6+44 \end{bmatrix}$$

$$= \begin{bmatrix} -15 & 32 & 24 \\ -35 & 64 & 38 \end{bmatrix} //$$

$$5) A = \begin{bmatrix} -8 & 0 & 5 \\ 4 & 19 & 6 \\ 18 & 90 & 37 \end{bmatrix} \quad B = \begin{bmatrix} -6 & 13 & 11 \\ 2 & -8 & 0 \\ 6 & 1 & 1 \end{bmatrix}$$

$$A \quad B = \begin{bmatrix} \cancel{(-8 \times -6)} + \cancel{(0 \times 2)} + \cancel{(5 \times 6)} & \cancel{(-8 \times 13)} + \cancel{(0 \times -8)} + \cancel{(5 \times 1)} & \cancel{(-8 \times 11)} + \cancel{(0 \times 0)} + \cancel{(5 \times 1)} \\ \cancel{(4 \times -6)} + \cancel{(19 \times 2)} + \cancel{(6 \times 6)} & \cancel{(4 \times 13)} + \cancel{(19 \times -8)} + \cancel{(6 \times 1)} & \cancel{(4 \times 11)} + \cancel{(19 \times 0)} + \cancel{(6 \times 1)} \\ \cancel{(18 \times -6)} + \cancel{(90 \times 2)} + \cancel{(37 \times 6)} & \cancel{(18 \times 13)} + \cancel{(90 \times -8)} + \cancel{(37 \times 1)} & \cancel{(18 \times 11)} + \cancel{(90 \times 0)} + \cancel{(37 \times 1)} \end{bmatrix}$$

$$AB = \begin{bmatrix} (-8 \times -6) + (0 \times 2) + (5 \times 6) & (-8 \times 13) + (0 \times -8) + (5 \times 1) & (-8 \times 11) + (0 \times 0) + (5 \times 1) \\ (4 \times -6) + (19 \times 2) + (6 \times 6) & (4 \times 13) + (19 \times -8) + (6 \times 1) & (4 \times 11) + (19 \times 0) + (6 \times 1) \\ (18 \times -6) + (90 \times 2) + (37 \times 6) & (18 \times 13) + (90 \times -8) + (37 \times 1) & (18 \times 11) + (90 \times 0) + (37 \times 1) \end{bmatrix}$$

$$= \begin{bmatrix} 48+0+30 & -104+0+5 & -88+0+5 \\ -24+38+36 & 52-152+6 & 44+0+6 \\ \cancel{108}+180+222 & 234-720+37 & 198+0+37 \end{bmatrix}$$

$$= \begin{bmatrix} 78 & -99 & -83 \\ 50 & -94 & 50 \\ \cancel{530} & -449 & 235 \end{bmatrix} //$$

$$(6 \times -8) + (1 \times 4) + (1 \times 8) \quad (6 \times 0) + (1 \times 19) + (1 \times 90) \quad (6 \times 5) + (1 \times 6) + (1 \times 10)$$

$$\begin{bmatrix} 48 + 52 + 198 & 0 + 247 + 990 & -30 + 78 + 407 \\ -16 - 32 + 0 & 0 - 152 + 0 & 10 - 48 + 0 \\ -48 + 4 + 18 & 0 + 19 + 90 & 30 + 6 + 37 \end{bmatrix}$$

$$\begin{bmatrix} 298 & 1237 & 455 \\ -48 & -152 & -38 \\ -26 & 109 & 73 \end{bmatrix}$$

and $AB \neq BA$ if $A = \begin{bmatrix} -4 \\ 17 \\ 116 \\ 42 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}$

$$B \underset{1 \times 3}{A} = \begin{bmatrix} -4 \times 1 & -4 \times 3 & -4 \times 2 \\ 17 \times 1 & 17 \times 3 & 17 \times 2 \\ 116 \times 1 & 116 \times 3 & 116 \times 2 \\ 42 \times 1 & 42 \times 3 & 42 \times 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} -4 & -12 & -8 \\ 17 & 51 & 34 \\ 116 & 348 & 232 \\ 42 & 126 & 84 \end{bmatrix}$$

Verify whether $AB = BA$

Sol: $A \underset{3 \times 3}{B} = \begin{bmatrix} (2 \times 1) + (3 \times 1) + (4 \times 0) & (2 \times 3) + (3 \times 2) + (4 \times 0) & (2 \times 0) + (3 \times 1) + (4 \times 2) \\ (1 \times 1) + (2 \times 1) + (3 \times 0) & (1 \times 3) + (2 \times 2) + (3 \times 0) & (1 \times 0) + (2 \times 1) + (3 \times 2) \\ (1 \times 1) + (1 \times 1) + (1 \times 0) & (1 \times 3) + (1 \times 2) + (1 \times 0) & (1 \times 0) + (1 \times 1) + (1 \times 2) \end{bmatrix}$

$$= \begin{bmatrix} 2+3+0 & 6+6+0 & 0+3+8 \\ 1+2+0 & 3+4+0 & 0+2+6 \\ 1+1+0 & 3+2+0 & 0+1+2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 5 & 12 & 11 \\ 3 & 7 & 8 \\ 2 & 5 & 3 \end{bmatrix}$$

$$B \underset{3 \times 3}{A} = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1 \times 2) + (3 \times 1) + (0 \times -1) & (1 \times 3) + (3 \times 2) + (0 \times 1) & (1 \times 4) + (3 \times 3) + (0 \times 1) \\ (1 \times 2) + (2 \times 1) + (1 \times 0) & (1 \times 3) + (2 \times 2) + (1 \times 1) & (1 \times 4) + (2 \times 3) + (1 \times 1) \\ (0 \times 2) + (0 \times 1) + (2 \times -1) & (0 \times 3) + (0 \times 2) + (2 \times 1) & (0 \times 4) + (0 \times 3) + (2 \times 1) \end{bmatrix}$$

$$= \begin{bmatrix} 2+3+0 & 3+6+0 & 4+9+0 \\ 2+2+1 & 3+4+1 & 4+6+1 \\ 0+0+2 & 0+0+2 & 0+0+2 \end{bmatrix}$$

$$BA = \begin{bmatrix} 5 & 9 & 13 \\ 5 & 8 & 11 \\ 2 & 2 & 2 \end{bmatrix}$$

$\therefore AB \neq BA$

8. If $A = \begin{bmatrix} -3 & 5 \\ 18 & 6 \end{bmatrix}$. Verify $A^2 - 2A + 5I = 0$ //

Sol: $A^2 = A \times A$.

$$A^2 = \begin{bmatrix} -3 & 5 \\ 18 & 6 \end{bmatrix} \begin{bmatrix} -3 & 5 \\ 18 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} (-3 \times -3) + (5 \times 18) & (-3 \times 5) + (5 \times 6) \\ (18 \times -3) + (6 \times 18) & (18 \times 5) + (6 \times 6) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 9+90 & -15+30 \\ -54+108 & 90+36 \end{bmatrix} = \begin{bmatrix} 99 & 15 \\ 54 & 126 \end{bmatrix} //$$

$$2A = 2 \begin{bmatrix} -3 & 5 \\ 18 & 6 \end{bmatrix} = \begin{bmatrix} -6 & 10 \\ 36 & 12 \end{bmatrix} //$$

$$5I = 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

Now,

$$A^2 - 2A + 5I = \begin{bmatrix} 99 & 15 \\ 54 & 126 \end{bmatrix} - \begin{bmatrix} -6 & 10 \\ 36 & 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 99 - (-6) & 15 - 10 \\ 54 - 36 & 126 - 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 105 & 5 \\ 18 & 114 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 110 & 5 \\ 18 & 119 \end{bmatrix} //$$

$\therefore A^2 - 2A + 5I \neq 0$ //

9. If $A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ Find $A^2 - 5A + 9I$

10. Prove that $A^2 - 4A - 5I = 0$ if $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

Sol:

$$9. A^2 = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} (2 \times 2) + (1 \times -1) + (1 \times 1) & (2 \times 1) + (1 \times 0) + (1 \times 3) & (2 \times 1) + (1 \times 1) + (1 \times -1) \\ (-1 \times 2) + (0 \times -1) + (1 \times 1) & (-1 \times 1) + (0 \times 0) + (1 \times 3) & (-1 \times 1) + (0 \times 1) + (1 \times -1) \\ (1 \times 2) + (3 \times -1) + (-1 \times 1) & (1 \times 1) + (3 \times 0) + (-1 \times 3) & (1 \times 1) + (3 \times 1) + (-1 \times -1) \end{bmatrix}$$

$$= \begin{bmatrix} 4-1+1 & 2+0+3 & 2+1-1 \\ -2+0+1 & -1+0+3 & -1+0-1 \\ 2-3-1 & 1+0-3 & 1+3+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & 5 & 2 \\ -1 & 2 & -2 \\ -2 & -2 & 5 \end{bmatrix} //$$

$$5A = 5 \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 3 & -1 \end{bmatrix} = \begin{bmatrix} 10 & 5 & 5 \\ -5 & 0 & 5 \\ 5 & 15 & -5 \end{bmatrix}$$

$$9I = 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} (1 \times 1) + (2 \times 2) + (2 \times 2) & (1 \times 2) + (2 \times 1) + (2 \times 2) & (1 \times 2) + (2 \times 2) + (2 \times 1) \\ (2 \times 1) + (1 \times 2) + (2 \times 2) & (2 \times 2) + (1 \times 1) + (2 \times 2) & (2 \times 2) + (1 \times 2) + (2 \times 1) \\ (2 \times 1) + (2 \times 2) + (1 \times 2) & (2 \times 2) + (2 \times 1) + (1 \times 2) & (2 \times 2) + (2 \times 2) + (1 \times 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

$$4A = 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} //$$

$$\therefore A^2 - 4A - 5I = 0 //$$

Hence proved.

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(3 \times 3) + (1 \times 4) + (1 \times 3) \quad (3 \times 6) + (1 \times -8) + (1 \times 1) \quad (3 \times 9) + (1 \times 4) + (1 \times 1)$$

$+24+57$	$18-48+19$	$57+84+19$
$2-32+42$	$24+64+14$	$76-112+14$
$9+4+3$	$18-8+1$	$57+14+1$

$$\begin{bmatrix} 90 & -11 & 160 \\ 22 & 102 & -22 \\ 16 & 11 & 72 \end{bmatrix} //$$

$$A = 6 \begin{bmatrix} 3 & 6 & 19 \\ 4 & -8 & 14 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 18 & 36 & 114 \\ 24 & -48 & 84 \\ 18 & 6 & 6 \end{bmatrix}$$

$$34I = 34 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 184 & -106 \\ -2 & 5 & 100 \end{bmatrix} \quad \text{pi}$$

[illegible]

Solve AB & BA

$$A = \begin{bmatrix} 1 & 9 & 16 \\ 4 & -5 & 18 \\ 1 & 1 & 2 \end{bmatrix}$$

Since there's no B matrix.

We cannot perform AB or BA //

Elementary row operations:

- The elementary row operations of a matrix involve
- i. Interchange of any two rows.
- ii. Multiplication of any one of the rows by a non-zero scalar.
- iii. Addition or subtraction of any 2 rows by which any one of the rows can be multiplied by non-zero scalar.

Symbolically

- Interchange $R_i \leftrightarrow R_j$
- Multiplication $R_i \rightarrow KR_i$
- Addition and subtraction $R_i \rightarrow R_i + KR_j$

Rank of a matrix:

The rank of a matrix is the no. of non-zero rows in the row reduced Echelon form (RREF)

Row Reduced Echelon form (RREF)

The matrix A is said to be in row reduced echelon form if

- The leading entry of each row is unity i.e., (one)
- All the entries below the leading entry are zero.
- The no. of zeroes appear in below the leading entry in each row must be greater than that appearing in the previous row.

Find rank of a matrix

$$A = \begin{bmatrix} 1 & 4 & -7 \\ 2 & -8 & 9 \\ -5 & 0 & 3 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

$$\begin{array}{lcl}
 R_1 \rightarrow R_1 - 2R_2 & & \\
 2 \rightarrow 2 - 2(1) = 0 & & \\
 -8 \rightarrow -8 - 2(4) = -16 & & \\
 9 \rightarrow 9 - 2(7) = 23 & & \\
 R_2 \rightarrow R_2 + 5R_1 & & \\
 -5 \rightarrow -5 + 5(1) = 0 & & \\
 0 \rightarrow 0 + 5(4) = 20 & & \\
 3 \rightarrow 3 + 5(-7) = -32 & &
 \end{array}$$

Now,

$$A = \begin{bmatrix} 1 & 4 & -7 \\ 0 & -16 & 23 \\ 0 & 20 & -32 \end{bmatrix} \begin{array}{l} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{array}$$

$$R_2 \rightarrow \frac{1}{-16} R_2$$

$$0 \rightarrow \frac{1}{-16} (0) = 0$$

$$-16 \rightarrow \frac{1}{-16} (-16) = 1$$

$$23 \rightarrow \frac{1}{-16} (23) = -\frac{23}{16}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 4 & -7 \\ 0 & 1 & -\frac{23}{16} \\ 0 & 20 & -32 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 20R_2$$

$$0 \rightarrow 0 - 20(0) = 0$$

$$20 \rightarrow 20 - 20(1) = 0$$

$$-32 \rightarrow -32 - 20(-\frac{23}{16}) = -3.25$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & -7 \\ 0 & 1 & -\frac{23}{16} \\ 0 & 0 & -3.25 \end{bmatrix}$$

$$\begin{array}{lcl}
 R_3 \rightarrow R_3 - 20R_2 & & \\
 0 \rightarrow 0 - 20(0) = 0 & & \\
 20 \rightarrow 20 - 20(1) = 0 & & \\
 -3 \rightarrow -3 - 20(-\frac{23}{16}) = 16 & &
 \end{array}$$

$$R_3 \rightarrow \frac{1}{-3.25} R_3$$

$$-3.25 \rightarrow \frac{1}{-3.25} (-3.25) = 1$$

$$\Rightarrow A = \begin{bmatrix} 1 & 4 & -7 \\ 0 & 1 & -\frac{23}{16} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\rho(A) = 3 //$$

$$2. A = \begin{bmatrix} 1 & -1 & 3 & 2 \\ 1 & 3 & -3 & 4 \\ 5 & 3 & 3 & 9 \end{bmatrix} \begin{array}{l} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{array}$$

Sol:

$$R_2 \rightarrow R_2 - R_1$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$3 \rightarrow 3 - 1(-1) = 4$$

$$-3 \rightarrow -3 - 1(3) = -6$$

$$4 \rightarrow 4 - 1(2) = 2$$

$$R_3 \rightarrow R_3 - 5R_1$$

$$5 \rightarrow 5 - 5(1) = 0$$

$$3 \rightarrow 3 - 5(-1) = 8$$

$$3 \rightarrow 3 - 5(3) = -12$$

$$9 \rightarrow 9 - 5(2) = -1$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 3 & 2 \\ 0 & 4 & -6 & 2 \\ 0 & 8 & -12 & -1 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{4} R_2 \rightarrow 4 \rightarrow \frac{1}{4} (4) = 1$$

$$R_2 \rightarrow \frac{1}{-6} R_2 \rightarrow -6 \rightarrow \frac{1}{-6} (-6) = 1$$

$$R_2 \rightarrow \frac{1}{-6} R_2 \rightarrow -6 \rightarrow \frac{1}{-6} (-6) = 1$$

$$3. A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 3 & 5 \\ 3 & 2 & 6 & 7 \end{bmatrix}$$

$$\text{Sol: } R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$2 \rightarrow 2 - 2(1) = 0$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$4 \rightarrow 4 - 2(2) = 0$$

$$2 \rightarrow 2 - 3(2) = -4$$

$$3 \rightarrow 3 - 2(1) = 1$$

$$6 \rightarrow 6 - 3(1) = 3$$

$$5 \rightarrow 5 - 2(4) = -3$$

$$7 \rightarrow 7 - 3(4) = -5$$

$$\Rightarrow A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & -3 \\ 0 & -4 & 3 & -5 \end{bmatrix}$$

Now, Interchange R_2 & R_3

$$A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & -4 & 3 & -5 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{-4} R_2$$

$$0 \rightarrow \frac{1}{-4} (0) = 0$$

$$-4 \rightarrow \frac{1}{-4} (-4) = 1$$

$$3 \rightarrow \frac{1}{-4} (3) = 3/-4$$

$$-5 \rightarrow \frac{1}{-4} (-5) = -5/-4$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \end{bmatrix} \quad \text{then interchange}$$

$$R_4 \rightarrow R_4 - 1R_2$$

$$0 \rightarrow 0 - 1(0) = 0$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$-1 \rightarrow -1 - 1(-1) = 0$$

$$\Rightarrow A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank}(A) = 2$$

5. find rank of $A = \begin{bmatrix} 3 & 2 & 2 & 4 \\ 5 & -8 & -4 & 2 \end{bmatrix}$

sol: $R_2 \rightarrow \frac{1}{5} R_2$

$2 \rightarrow \frac{1}{2} (2) = 1$

$R_2 \rightarrow R_2 - 1 \rightarrow \frac{1}{2} (1) = \frac{1}{2}$

$5 \rightarrow \frac{1}{2} (5) = \frac{5}{2}$

$4 \rightarrow \frac{1}{2} (4) = 2$

② $A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 3 & 2 & 2 & 4 \\ 5 & -8 & -4 & 2 \end{bmatrix}$

② $R_2 \rightarrow R_2 - 3R_1$

$3 \rightarrow 3 - 3(1) = 0$

$2 \rightarrow 2 - 3(2) = -4$

$2 \rightarrow 2 - 3(2) = -4$

$4 \rightarrow 4 - 3(2) = -2$

$R_3 \rightarrow R_3 - 5R_1$

$5 \rightarrow 5 - 5(1) = 0$

$-8 \rightarrow -8 - 5(2) = -18$

$-4 \rightarrow -4 - 5(2) = -14$

$2 \rightarrow 2 - 5(2) = -8$

$\begin{bmatrix} 1 & 0.5 & 5.5 & 2 \\ 0 & -4 & -14 & -2 \\ 0 & -10.5 & -16.5 & -8 \end{bmatrix}$

Now, $R_2 \rightarrow \frac{1}{-4} R_2$, $0 \rightarrow \frac{1}{-4} (0) = 0$

$0.5 \rightarrow \frac{1}{0.5} (0.5) = 1$

$-5.5 \rightarrow \frac{1}{0.5} (-5.5) = -11$

$-2 \rightarrow \frac{1}{0.5} (-2) = -4$

$\begin{bmatrix} 1 & 0.5 & 5.5 & 2 \\ 0 & 1 & -11 & -4 \\ 0 & -10.5 & -16.5 & -8 \end{bmatrix}$

$R_3 \rightarrow R_3 + 10.5(R_2)$

$0 \rightarrow 0 + 10.5(0) = 0$

$-10.5 \rightarrow -10.5 + 10.5(1) = 0$

$-16.5 \rightarrow -16.5 + 10.5(-11) = -17$

$-8 \rightarrow -8 + 10.5(-4) = -50$

$\Rightarrow \begin{bmatrix} 1 & 0.5 & 5.5 & 2 \\ 0 & 1 & -11 & -4 \\ 0 & 0 & -17 & -50 \end{bmatrix}$

$R_2 \rightarrow \frac{1}{-17} R_2$

$0 \rightarrow \frac{1}{-17} (0) = 0$, $-17 \rightarrow \frac{1}{-17} (-17) = 1$

$0 \rightarrow \frac{1}{-17} (0) = 0$, $-50 \rightarrow \frac{1}{-17} (-50) = 2.94$

$\Rightarrow \begin{bmatrix} 1 & 0.5 & 5.5 & 2 \\ 0 & 1 & -11 & -4 \\ 0 & 0 & 1 & 2.94 \end{bmatrix} \therefore \text{Rank } S(A) = 3$

Final rank of A = $\begin{bmatrix} -8 & 1 & 3 & 2 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$

Rank $\rho(A)$
 $R_1 \rightarrow \frac{1}{8}(R_1)$
 $8 \rightarrow \frac{1}{8}(8) = 1$

$1 \rightarrow \frac{1}{8}(1) = 0.125$

$3 \rightarrow \frac{1}{8}(3) = 0.375$

$6 \rightarrow \frac{1}{8}(6) = 0.75$

$\begin{bmatrix} 1 & 0.125 & 0.375 & 0.75 \\ 0 & 3 & 2 & 2 \\ 8 & -1 & -3 & 4 \end{bmatrix}$

$\rightarrow R_3 + 8(R_1)$

$8 \rightarrow -8 + 8(1) = 0$

$1 \rightarrow -1 + 8(0.125) = 0$

$3 \rightarrow -3 + 8(0.375) = 0$

$4 \rightarrow 4 + 8(0.75) = 10$

$\begin{bmatrix} 1 & 0.125 & 0.375 & 0.75 \\ 0 & 3 & 2 & 2 \\ 0 & 0 & 0 & 10 \end{bmatrix}$

exchange R_2 and R_3
 $R_2 \rightarrow \frac{1}{10}(R_2)$

$10 \rightarrow \frac{1}{10}(10) = 1$

$3 \rightarrow \frac{1}{3}(3) = 1$

$2 \rightarrow \frac{1}{3}(2) = 0.67$

$2 \rightarrow \frac{1}{3}(2) = 0.67$

$\Rightarrow \begin{bmatrix} 1 & 0.125 & 0.375 & 0.75 \\ 0 & 1 & 0.67 & 0.67 \\ 0 & 0 & 0 & 10 \end{bmatrix}$

$R_3 \rightarrow \frac{1}{10}(R_3)$

$0 \rightarrow \frac{1}{10}(0) = 0$

$0 \rightarrow \frac{1}{10}(0) = 0$

$0 \rightarrow \frac{1}{10}(0) = 0$

$10 \rightarrow \frac{1}{10}(10) = 1$

$\Rightarrow \begin{bmatrix} 1 & 0.125 & 0.375 & 0.75 \\ 0 & 1 & 0.67 & 0.67 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Rank $\rho(A) = 3$

$$R_1 \rightarrow \frac{1}{2} (R_1)$$

$$2 \rightarrow \frac{1}{2} (2) = 1$$

$$3 \rightarrow \frac{1}{2} (3) = 1.5$$

$$-1 \rightarrow \frac{1}{2} (-1) = -0.5$$

$$-1 \rightarrow \frac{1}{2} (-1) = -0.5$$

$$\Rightarrow \begin{bmatrix} 1 & 1.5 & -0.5 & -0.5 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 1(R_1)$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$-1 \rightarrow -1 - 1(1.5) = -2.5$$

$$-2 \rightarrow -2 - 1(-0.5) = -1.5$$

$$-4 \rightarrow -4 - 1(-0.5) = -3.5$$

$$R_3 \rightarrow R_3 - 3(R_1)$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$1 \rightarrow 1 - 3(1.5) = -3.5$$

$$3 \rightarrow 3 - 3(-0.5) = 4.5$$

$$-2 \rightarrow -2 - 3(-0.5) = -0.5$$

$$R_4 \rightarrow R_4 - 6(R_1)$$

$$6 \rightarrow 6 - 6(1) = 0$$

$$3 \rightarrow 3 - 6(1.5) = -6$$

$$0 \rightarrow 0 - 6(-0.5) = 3$$

$$-7 \rightarrow -7 - 6(-0.5) = -4$$

$$\Rightarrow \begin{bmatrix} 1 & 1.5 & -0.5 & -0.5 \\ 0 & 1 & 0.6 & 1.4 \\ 0 & 0 & 6.6 & 4.4 \\ 0 & 0 & 6.6 & 4.4 \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{6.6} (R_3)$$

$$0 \rightarrow \frac{1}{6.6} (0) = 0$$

$$0 \rightarrow \frac{1}{6.6} (0) = 0$$

$$6.6 \rightarrow \frac{1}{6.6} (6.6) = 1$$

$$4.4 \rightarrow \frac{1}{6.6} (4.4) = 0.67$$

$$\begin{bmatrix} 1 & 1.5 & -0.5 & -0.5 \\ 0 & 1 & 0.6 & 1.4 \\ 0 & 0 & 1 & 0.67 \\ 0 & 0 & 6.6 & 4.4 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 6.6(R_3)$$

$$6.6 \rightarrow 6.6 - 6.6(1) = 0$$

$$4.4 \rightarrow 4.4 - 6.6(0.67) = 0$$

$$\Rightarrow 4.4 - 4.4 = 0$$

0 //

$$R_4 \rightarrow \frac{1}{6.6} (R_4)$$

$$0 \rightarrow \frac{1}{6.6} (0) = 0$$

$$0 \rightarrow \frac{1}{6.6} (0) = 0$$

$$6.6 \rightarrow \frac{1}{6.6} (6.6) = 1$$

$$4.4 \rightarrow \frac{1}{6.6} (4.4) = 0.67$$

$$\text{Rank } S(A) = 3 //$$

$$\Rightarrow \begin{bmatrix} 1 & 1.5 & -0.5 & -0.5 \\ 0 & 1 & 0.6 & 1.4 \\ 0 & 0 & 1 & 0.67 \\ 0 & 0 & 1 & 0.67 \end{bmatrix}$$

$$\text{Rank } S(A) = 4 //$$

$$8. A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & 3 & 2 & 1 \\ 2 & 0 & 3 & 2 \\ -3 & 3 & 3 & 3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 1(R_1)$$

$$R_3 \rightarrow R_3 - 2(R_1)$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$2 \rightarrow 2 - 2(1) = 0$$

$$3 \rightarrow 3 - 1(1) = 2$$

$$0 \rightarrow 0 - 2(1) = -2$$

$$2 \rightarrow 2 - 1(-1) = 3$$

$$3 \rightarrow 3 - 2(-1) = 5$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$2 \rightarrow 2 - 2(1) = 0$$

$$R_4 \rightarrow R_4 + 3(R_1)$$

$$-3 \rightarrow -3 + 3(1) = 0$$

$$3 \rightarrow 3 + 3(1) = 6$$

$$3 \rightarrow 3 + 3(-1) = 0$$

$$3 \rightarrow 3 + 3(1) = 6$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 2 & 3 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 6 & 0 & 6 \end{bmatrix}$$

$$\frac{1}{2} \rightarrow \frac{1}{2}$$

$$\frac{1}{2} (R_2)$$

$$1(0) = 0$$

$$1(2) = 1$$

$$1(3) = 1.5$$

$$1(0) = 0$$

$$\begin{array}{c|c|c} 1 & -1 & 1 \\ 1 & 1.5 & 0 \\ -2 & 5 & 0 \\ 6 & 0 & 6 \end{array} \quad \begin{array}{l} R_1 \rightarrow R_1 - R_2 \\ R_2 \rightarrow R_2 - 6(R_1) \\ R_3 \rightarrow R_3 - 5(R_1) \\ R_4 \rightarrow R_4 - 6(R_1) \end{array}$$

$$R_2 \rightarrow R_2 - 6(R_1)$$

$$0 \rightarrow 0 - 6(0) = 0$$

$$6 \rightarrow 6 - 6(1) = 0$$

$$0 \rightarrow 0 - 6(1.5) = 0$$

$$6 \rightarrow 6 - 6(0) = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1.5 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$0 \rightarrow \frac{1}{8} (0) = 0$$

$$8 \rightarrow \frac{1}{8} (8) = 1$$

$$0 \rightarrow \frac{1}{8} (0) = 0$$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & 1.5 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} //$$

$$\text{Rank}(A) = 3 //$$

$$9. A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & 0 & 1 & 1 \\ 3 & 3 & 2 & 1 & 1 \end{bmatrix}$$

Sol: Interchange R_1 and R_2

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 1 & 1 & 0 & 1 & 1 \\ 3 & 3 & 2 & 1 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 1(R_2)$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$1 \rightarrow 1 - 1(1) = 0$$

$$0 \rightarrow 0 - 1(2) = -2$$

$$1 \rightarrow 1 - 1(3) = -2$$

$$1 \rightarrow 1 - 1(3) = -2$$

$$R_2 \rightarrow R_2 - 3(R_1)$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$2 \rightarrow 2 - 3(2) = -4$$

$$1 \rightarrow 1 - 3(3) = -8$$

$$1 \rightarrow 1 - 3(3) = -8$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & -2 \\ 0 & 9 & -11 \\ 3 & 1 & 0 \\ 2 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow R - 2(R_1)$$

LHS:

$$2 \rightarrow 2 - 2(1) = 0$$

$$5 \rightarrow 5 - 2(3) = -1$$

$$0 \rightarrow 0 - 2(-2) = -4$$

RHS:

$$0 \rightarrow 0 - 2(1) = -2$$

$$0 \rightarrow 0 - 2(0) = 0$$

$$1 \rightarrow 1 - 2(0) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & -2 \\ 0 & 9 & -11 \\ 0 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow \frac{1}{9}(R_2)$$

LHS:

$$0 \rightarrow \frac{1}{9}(0) = 0$$

$$9 \rightarrow \frac{1}{9}(9) = 1$$

$$-11 \rightarrow \frac{1}{9}(-11) = -1.22$$

RHS:

$$3 \rightarrow \frac{1}{9}(3) = 0.33$$

$$1 \rightarrow \frac{1}{9}(1) = 0.11$$

$$0 \rightarrow \frac{1}{9}(0) = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -1.22 \\ 0 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.33 & 0.11 & 0 \\ -2 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow R + 1(R_2), \text{LHS:}$$

$$0 \rightarrow 0 + 1(0) = 0$$

$$-1 \rightarrow -1 + 1(1) = 0$$

$$-4 \rightarrow -4 + 1(-1.22) = -5.22$$

RHS:

$$-2 \rightarrow -2 + 1(0.33) = -1.67$$

$$0 \rightarrow 0 + 1(0.11) = 0.11$$

$$1 \rightarrow 1 + 1(0) = 1$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 1 & 4 & 7 & 10 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 1(R_1) & R_3 &\rightarrow R_3 - 1(R_1) \\ 1 &\rightarrow 1 - 1(1) = 0 & 1 &\rightarrow 1 - 1(1) = 0 \\ 2 &\rightarrow 2 - 1(1) = 1 & 3 &\rightarrow 3 - 1(1) = 2 \\ 3 &\rightarrow 3 - 1(1) = 2 & 5 &\rightarrow 5 - 1(1) = 4 \\ 4 &\rightarrow 4 - 1(1) = 3 & 7 &\rightarrow 7 - 1(1) = 6 \end{aligned}$$

$$\begin{aligned} R_4 &\rightarrow R_4 - 1(R_1) \\ 1 &\rightarrow 1 - 1(1) = 0 \\ 4 &\rightarrow 4 - 1(1) = 3 \\ 7 &\rightarrow 7 - 1(1) = 6 \\ 10 &\rightarrow 10 - 1(1) = 9 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \\ 0 & 3 & 6 & 9 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 2(R_1) & R_3 &\rightarrow R_3 - 3(R_1) \\ 3 &\rightarrow 3 - 2(1) = 1 & 3 &\rightarrow 3 - 3(1) = 0 \\ 0 &\rightarrow 0 - 2(0) = 0 & 0 &\rightarrow 0 - 3(0) = 0 \\ 2 &\rightarrow 2 - 2(1) = 0 & 3 &\rightarrow 3 - 3(1) = 0 \\ 4 &\rightarrow 4 - 2(2) = 0 & 6 &\rightarrow 6 - 3(2) = 0 \\ 6 &\rightarrow 6 - 2(3) = 0 & 9 &\rightarrow 9 - 3(3) = 0 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank } S(A) = 2 //$$

* Inverse of a matrix by elementary row operations
If A is a square matrix of order n and if there exists another matrix B of the same order such that $AB = BA = I$

$$\text{then } B = A^{-1}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

V.K.
(12m)

Q. Find Inverse of the following matrix

$$1. A = \begin{bmatrix} 2 & -6 \\ 1 & -2 \end{bmatrix}$$

$$\text{Sol: Consider } A = IA \quad \rightarrow I = A^{-1}A$$

$$\begin{bmatrix} 2 & -6 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$\begin{aligned} R_2 &\leftrightarrow R_1 \\ \begin{bmatrix} 1 & -2 \\ 2 & -6 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} A \end{aligned}$$

$$R_2 \rightarrow R_2 - 2(R_1)$$

LHS:

$$\begin{aligned} 2 &\rightarrow 2 - 2(1) = 0 \\ -6 &\rightarrow -6 - 2(-2) = -2 \end{aligned}$$

RHS:

$$\begin{aligned} 1 &\rightarrow 1 - 2(0) = 1 \\ 0 &\rightarrow 0 - 2(1) = -2 \end{aligned}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & -2 & -2 & -2 \\ 0 & 0 & -4 & -8 & -8 \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{-2} (R_3)$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$-2 \rightarrow \frac{1}{-2} (-2) = 1$$

$$-2 \rightarrow \frac{1}{-2} (-2) = 1$$

$$-2 \rightarrow \frac{1}{-2} (-2) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -4 & -8 & -8 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + 4(R_3)$$

$$0 \rightarrow 0 + 4(0) = 0$$

$$0 \rightarrow 0 + 4(0) = 0$$

$$-4 \rightarrow -4 + 4(1) = 0$$

$$-8 \rightarrow -8 + 4(1) = -4$$

$$-8 \rightarrow -8 + 4(1) = -4$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 4 & 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -4 & -4 \end{bmatrix}$$

$$R_4 \rightarrow \frac{1}{-4} (R_4)$$

$$0 \rightarrow \frac{1}{-4} (0) = 0$$

$$0 \rightarrow \frac{1}{-4} (0) = 0$$

$$0 \rightarrow \frac{1}{-4} (0) = 0$$

$$-4 \rightarrow \frac{1}{-4} (-4) = 1$$

$$-4 \rightarrow \frac{1}{-4} (-4) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{Rank } P(A) = 4$$

10.

$$\begin{bmatrix} -2 \\ -1.22 \\ -5.22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.33 & 0.11 & 0 \\ -1.67 & 0.11 & 1 \end{bmatrix} \cdot A$$

(R₃)

22

$$(0) = 0$$

22

$$(0) = 0$$

5.22

$$\rightarrow 1 \cdot (-5.22) = 1$$

RHS:

$$-1.67 \rightarrow \frac{1}{-5.22} (-1.67) = 0.32$$

$$0.11 \rightarrow \frac{1}{-5.22} (0.11) = 0.02$$

$$1 \rightarrow \frac{1}{-5.22} (1) = -0.19$$

$$\begin{bmatrix} 3 & -2 \\ 1 & -1.22 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.33 & 0.11 & 0 \\ 0.32 & -0.02 & -0.19 \end{bmatrix} \cdot A$$

$$\rightarrow R_2 + 1(R_3) - 1.22(R_1)$$

RHS:

$$\rightarrow 0 + 1.22(0) = 0 \quad 0.33 \rightarrow 0.33 + 1.22(0.32) = 0.72$$

$$\rightarrow 1 + 1.22(0) = 1 \quad 0.11 \rightarrow 0.11 + 1.22(-0.02) = 0.09$$

$$22 \rightarrow -1.22 + 1.22(1) = 0 \quad 0 \rightarrow 0 + 1.22(-0.19) = -0.23$$

$$\begin{bmatrix} 3 & -2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.72 & 0.09 & 0.23 \\ 0.32 & -0.02 & -0.19 \end{bmatrix} \cdot A$$

$$\rightarrow R_1 + 2(R_3)$$

RHS:

$$\rightarrow 1 + 2(0) = 1 \quad 1 \rightarrow 1 + 2(0.32) = 1.64$$

$$\rightarrow 3 + 2(0) = 3 \quad 0 \rightarrow 0 + 2(-0.02) = -0.04$$

$$2 \rightarrow -2 + 2(1) = 0 \quad 0 \rightarrow 0 + 2(-0.19) = -0.38$$

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.64 & -0.04 & -0.38 \\ 0.72 & 0.09 & 0.23 \\ 0.32 & -0.02 & -0.19 \end{bmatrix} \cdot A$$

$$R \rightarrow R - 3(R_2)$$

LHS:

$$1 \rightarrow 1 - 3(0) = 1$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$0 \rightarrow 0 - 3(0) = 0$$

RHS:

$$1.64 \rightarrow 1.64 - 3(0.72) = -0.52$$

$$-0.04 \rightarrow -0.04 - 3(0.09) = -0.31$$

$$-0.38 \rightarrow -0.38 - 3(0.23) = -1.07$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.52 & -0.31 & -1.07 \\ 0.72 & 0.09 & 0.23 \\ 0.32 & -0.02 & -0.19 \end{bmatrix} \cdot A$$

$$A' = \begin{bmatrix} -0.52 & -0.31 & -1.07 \\ 0.72 & 0.09 & 0.23 \\ 0.32 & -0.02 & -0.19 \end{bmatrix}$$

$$3. \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Sol: Consider $A = IA$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - 4(R_1)$$

LHS:

$$4 \rightarrow 4 - 4(1) = 0$$

$$5 \rightarrow 5 - 4(2) = -3$$

$$6 \rightarrow 6 - 4(3) = -6$$

RHS:

$$0 \rightarrow 0 - 4(1) = -4$$

$$1 \rightarrow 1 - 4(0) = 1$$

$$0 \rightarrow 0 - 4(0) = 0$$

$$R_3 \rightarrow R_3 - 7(R_1)$$

LHS:

$$7 \rightarrow 7 - 7(1) = 0$$

$$8 \rightarrow 8 - 7(2) = -6$$

$$9 \rightarrow 9 - 7(3) = -12$$

RHS:

$$0 \rightarrow 0 - 7(1) = -7$$

$$0 \rightarrow 0 - 7(0) = 0$$

$$1 \rightarrow 1 - 7(0) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & 6 \\ 0 & -6 & -12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ -7 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow \frac{1}{-3}(R_2)$$

LHS:

$$0 \rightarrow \frac{1}{-3}(0) = 0$$

$$-3 \rightarrow \frac{1}{-3}(-3) = 1$$

RHS:

$$-4 \rightarrow \frac{1}{-3}(-4) = 1.33$$

$$1 \rightarrow \frac{1}{-3}(1) = -0.33$$

$$6 \rightarrow \frac{1}{-3}(6) = -2 \quad \left| \quad 0 \rightarrow \frac{1}{-3}(0) = 0 \right.$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & -6 & -12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1.33 & -0.33 & 0 \\ -7 & 0 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + 6(R_2)$$

LHS:

$$0 \rightarrow 0 + 6(0) = 0$$

$$-6 \rightarrow -6 + 6(1) = 0$$

$$-12 \rightarrow -12 + 6(-2) = -24$$

RHS:

$$-7 \rightarrow -7 + 6(1.33) = 0.98$$

$$0 \rightarrow 0 + 6(-0.33) = -1.98$$

$$1 \rightarrow 1 + 6(0) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & -24 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1.33 & -0.33 & 0 \\ 0.98 & -1.98 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow \frac{1}{-24}(R_3)$$

LHS:

$$0 \rightarrow \frac{1}{-24}(0) = 0$$

$$0 \rightarrow \frac{1}{-24}(0) = 0$$

$$-24 \rightarrow \frac{1}{-24}(-24) = 1$$

RHS:

$$0.98 \rightarrow \frac{1}{-24}(0.98) = -0.04$$

$$-1.98 \rightarrow \frac{1}{-24}(-1.98) = 0.08$$

$$1 \rightarrow \frac{1}{-24}(1) = -0.04$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1.33 & -0.33 & 0 \\ -0.04 & 0.08 & 0.04 \end{bmatrix} A$$

$$R \rightarrow R_2 + 2(R_3)$$

$$\begin{aligned} \text{LHS:} \\ 0 &\rightarrow 0 + 2(0) = 0 \\ 1 &\rightarrow 1 + 2(0) = 1 \\ -2 &\rightarrow -2 + 2(1) = 0 \end{aligned}$$

$$\text{RHS:}$$

$$\begin{aligned} 1.33 &\rightarrow 1.33 + 2(-0.04) = 1.25 \\ -0.33 &\rightarrow -0.33 + 2(0.08) = -0.17 \\ 0 &\rightarrow 0 + 2(0.04) = 0.08 \end{aligned}$$

$$R \rightarrow R_1 - 3(R_3)$$

$$\begin{aligned} \text{LHS:} \\ 1 &\rightarrow 1 - 3(0) = 1 \\ 2 &\rightarrow 2 - 3(0) = 2 \\ 3 &\rightarrow 3 - 3(1) = 0 \end{aligned}$$

$$\text{RHS:}$$

$$\begin{aligned} 1 &\rightarrow 1 - 3(-0.04) = 1.12 \\ 0 &\rightarrow 0 - 3(0.08) = -0.24 \\ 0 &\rightarrow 0 - 3(0.04) = -0.12 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 2 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.12 & -0.24 & -0.12 \\ 1.25 & -0.17 & 0.08 \\ -0.04 & 0.08 & 0.04 \end{bmatrix} A$$

$$R \rightarrow R_1 - 2(R_2)$$

$$\begin{aligned} \text{LHS:} \\ 1 &\rightarrow 1 - 2(0) = 1 \\ 2 &\rightarrow 2 - 2(1) = 0 \\ 0 &\rightarrow 0 - 2(0) = 0 \end{aligned}$$

$$\text{RHS:}$$

$$\begin{aligned} 1.12 &\rightarrow 1.12 - 2(1.25) = -1.38 \\ -0.24 &\rightarrow -0.24 - 2(-0.17) = 0.1 \\ -0.12 &\rightarrow -0.12 - 2(0.08) = -0.28 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1.38 & 0.1 & -0.28 \\ 1.25 & -0.17 & 0.08 \\ -0.04 & 0.08 & 0.04 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} -1.38 & 0.1 & -0.28 \\ 1.25 & -0.17 & 0.08 \\ -0.04 & 0.08 & 0.04 \end{bmatrix}$$

$$4. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 5 & 7 \end{bmatrix}$$

$$\text{Sol: Consider } A = IA$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow$$

$$R \rightarrow R_2 - 2(R_1)$$

$$\text{LHS:}$$

$$\begin{aligned} 2 &\rightarrow 2 - 2(1) = 0 \\ 3 &\rightarrow 3 - 2(2) = -1 \\ 4 &\rightarrow 4 - 2(3) = -2 \end{aligned}$$

$$\text{RHS:}$$

$$\begin{aligned} 0 &\rightarrow 0 - 2(1) = -2 \\ 1 &\rightarrow 1 - 2(0) = 1 \\ 0 &\rightarrow 0 - 2(0) = 0 \end{aligned}$$

$$R \rightarrow R_3 - 1(R_1)$$

$$\text{LHS:}$$

$$\begin{aligned} 1 &\rightarrow 1 - 1(1) = 0 \\ 5 &\rightarrow 5 - 1(2) = 3 \\ 7 &\rightarrow 7 - 1(3) = 4 \end{aligned}$$

$$\text{RHS:}$$

$$\begin{aligned} 0 &\rightarrow 0 - 1(1) = -1 \\ 0 &\rightarrow 0 - 1(0) = 0 \\ 1 &\rightarrow 1 - 1(0) = 1 \end{aligned}$$

$$\Rightarrow$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow \frac{1}{-1}(R_2)$$

$$\text{LHS:}$$

$$\begin{aligned} 0 &\rightarrow \frac{1}{-1}(0) = 0 \\ -1 &\rightarrow \frac{1}{-1}(-1) = 1 \\ -1 &\rightarrow \frac{1}{-1}(-1) = 1 \end{aligned}$$

$$\text{RHS:}$$

$$\begin{aligned} -2 &\rightarrow \frac{1}{-1}(-2) = 2 \\ -1 &\rightarrow \frac{1}{-1}(-1) = 1 \\ -1 &\rightarrow \frac{1}{-1}(-1) = 1 \end{aligned}$$

$$\begin{array}{c} -2 \rightarrow 1 \quad (2) = 2 \\ -1 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow R - 3(R_2)$$

LHS:

RHS:

$$0 \rightarrow 0 - 3(0) = 0$$

$$-1 \rightarrow -1 - 3(2) = -7$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$0 \rightarrow 0 - 3(1) = -3$$

$$4 \rightarrow 4 - 3(2) = -2$$

$$1 \rightarrow 1 - 3(0) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -7 & -3 & 1 \end{bmatrix} A$$

$$R \rightarrow \frac{1}{-2} (R_3)$$

LHS:

RHS:

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$-7 \rightarrow \frac{1}{-2} (-7) = 3.5$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$-3 \rightarrow \frac{1}{-2} (-3) = 1.5$$

$$2 \rightarrow \frac{1}{-2} (-2) = 1$$

$$1 \rightarrow \frac{1}{-2} (1) = -0.5$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

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$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$0 \rightarrow \frac{1}{-2} (0) = 0$$

$$R \rightarrow R - 3(R_3)$$

LHS:

RHS:

$$1 \rightarrow 1 - 3(0) = 1$$

$$1 \rightarrow 1 - 3(3.5) = -9.5$$

$$2 \rightarrow 2 - 3(0) = 2$$

$$0 \rightarrow 0 - 3(1.5) = -4.5$$

$$3 \rightarrow 3 - 3(1) = 0$$

$$0 \rightarrow 0 - 3(-0.5) = 1.5$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -9.5 & -4.5 & 1.5 \\ -5 & -2 & 1 \\ 3.5 & 1.5 & -0.5 \end{bmatrix} A$$

$$R \rightarrow R - 2(R_2)$$

LHS:

RHS:

$$1 \rightarrow 1 - 2(0) = 1$$

$$-9.5 \rightarrow -9.5 - 2(-5) = 0.5$$

$$2 \rightarrow 2 - 2(1) = 0$$

$$-4.5 \rightarrow -4.5 - 2(-2) = -0.5$$

$$0 \rightarrow 0 - 2(0) = 0$$

$$1.5 \rightarrow 1.5 - 2(1) = -0.5$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 & -0.5 \\ -5 & -2 & 1 \\ 3.5 & 1.5 & -0.5 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} 0.5 & -0.5 & -0.5 \\ -5 & -2 & 1 \\ 3.5 & 1.5 & -0.5 \end{bmatrix}$$

Find Rank & Inverse

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 3 & 1 \\ 0 & 5 & -2 \end{bmatrix}$$

Sol: Interchange $R_1 \rightarrow R_3$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 5 & -2 \\ 2 & 3 & 1 \end{bmatrix}$$

Consider $A = IA$

$$R_3 \rightarrow R_3 - 2(R_1)$$

LHS:

$$2 \rightarrow 2 - 2(1) = 0$$

$$3 \rightarrow 3 - 2(-2) = -1$$

$$1 \rightarrow 1 - 2(-1) = 3$$

RHS:

$$0 \rightarrow 0 - 2(1) = -2$$

$$0 \rightarrow 0 - 2(0) = 0$$

$$1 \rightarrow 1 - 2(0) = 1$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 5 & -2 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 5 & -2 \\ 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + 1(R_2)$$

LHS:

$$0 \rightarrow 0 + 1(0) = 0$$

$$5 \rightarrow 5 + 1(5) = 10$$

$$2 \rightarrow 2 + 1(-2) = 0$$

RHS:

$$0 \rightarrow 0 + 1(0) = 0$$

$$5 \rightarrow 5 + 1(5) = 10$$

$$-2 \rightarrow -2 + 1(-2) = -4$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + 1(R_2)$$

LHS:

$$0 \rightarrow 0 + 1(0) = 0$$

$$-1 \rightarrow -1 + 1(1) = 0$$

$$3 \rightarrow 3 + 1(-0.4) = 2.6$$

RHS:

$$-2 \rightarrow -2 + 1(0) = -2$$

$$0 \rightarrow 0 + 1(1) = 1$$

$$1 \rightarrow 1 + 1(-0.4) = 0.6$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 0 & 0 & 0.4 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 2 & 1 & 0.6 \end{bmatrix} A$$

$$R_3 \rightarrow \frac{1}{0.4}(R_3)$$

LHS:

$$0 \rightarrow 0 + 1(0) = 0$$

$$0 \rightarrow 0 + 1(0) = 0$$

$$0.4 \rightarrow 0.4 + 1(0.4) = 0.8$$

RHS:

$$2 \rightarrow 2 + 1(2) = 4$$

$$1 \rightarrow 1 + 1(1) = 2$$

$$0.6 \rightarrow 0.6 + 1(0.6) = 1.2$$

$$0.4 \rightarrow 0.4 + 1(0.4) = 0.8$$

$$0.4 \rightarrow 0.4 + 1(0.4) = 0.8$$

$$0.4 \rightarrow 0.4 + 1(0.4) = 0.8$$

$$0.4 \rightarrow 0.4 + 1(0.4) = 0.8$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 5 & 2.5 & 1.5 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 + 0.4(R_2)$$

LHS:

$$0 \rightarrow 0 + 0.4(0) = 0$$

$$1 \rightarrow 1 + 0.4(0) = 1$$

$$-0.4 \rightarrow -0.4 + 0.4(1) = 0$$

RHS:

$$0 \rightarrow 0 + 0.4(5) = 2$$

$$1 \rightarrow 1 + 0.4(2.5) = 2$$

$$-0.4 \rightarrow -0.4 + 0.4(1.5) = 0$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & -0.4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 0.2 \\ 5 & 2.5 & 1.5 \end{bmatrix} A$$

$$R \rightarrow R+1(R_1)$$

$$\text{LHS: } \begin{array}{l} R \rightarrow R+1(R_1) \\ 1 \rightarrow 1+1(0)=1 \\ -2 \rightarrow -2+1(0)=-2 \\ -1 \rightarrow -1+1(0)=-1 \end{array} \quad \text{RHS: } \begin{array}{l} 1 \rightarrow 1+1(5)=6 \\ -2 \rightarrow -2+1(2.5)=0.5 \\ -1 \rightarrow -1+1(1.5)=0.5 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 0.5 & 0.5 \\ 2 & 2 & 0.2 \\ 5 & 2.5 & 1.5 \end{bmatrix} A$$

$$R \rightarrow R+2(R_2)$$

$$\text{LHS: } \begin{array}{l} 1 \rightarrow 1+2(0)=1 \\ -2 \rightarrow -2+2(1)=0 \\ 0 \rightarrow 0+2(0)=0 \end{array} \quad \text{RHS: } \begin{array}{l} 6 \rightarrow 6+2(2)=10 \\ 0.5 \rightarrow 0.5+2(2)=4.5 \\ 0.5 \rightarrow 0.5+2(0.2)=0.9 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 4.5 & 0.9 \\ 2 & 2 & 0.2 \\ 5 & 2.5 & 1.5 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} 10 & 4.5 & 0.9 \\ 2 & 2 & 0.2 \\ 5 & 2.5 & 1.5 \end{bmatrix}$$

$$5. \quad A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

Sol: Interchange P_1 and P_2

$$\Rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

Consider $A = I A$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow R_3 - 1(R_1)$$

$$\text{LHS: } \begin{array}{l} 1 \rightarrow 1-1(1)=0 \\ 2 \rightarrow 2-1(0)=2 \\ 0 \rightarrow 0-1(2)=-2 \end{array} \quad \text{RHS: } \begin{array}{l} 0 \rightarrow 0-1(1)=-1 \\ 0 \rightarrow 0-1(0)=0 \\ 1 \rightarrow 1-1(0)=1 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A$$

$$R \rightarrow R_3 - 2(R_2)$$

$$\text{LHS: } \begin{array}{l} 0 \rightarrow 0-2(0)=0 \\ 2 \rightarrow 2-2(1)=0 \\ -2 \rightarrow -2-2(2)=-6 \end{array} \quad \text{RHS: } \begin{array}{l} -1 \rightarrow -1-2(0)=-1 \\ 0 \rightarrow 0-2(1)=-2 \\ 1 \rightarrow 1-2(0)=1 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & -6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -2 & 1 \end{bmatrix} A$$

System of Linear Equations:

A system of equations in which all the unknown quantities occur in the first degree is called system of linear equations.

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned} \right\} \text{①}$$

where $a_{11}, a_{12}, \dots, a_{1n}, b_1, b_2, \dots, b_m$ are the constants
is called a system of m equations & n unknown.

If in the above system of m equations ①

$$b_1 = b_2 = \dots = b_m = 0$$

$\therefore$ The system is called system of m -homogeneous equation in n -unknowns.

* A solution of linear equations is a set of values $x_1, x_2, \dots, x_n$ which satisfy all the n -equations simultaneously.

⑧ $\therefore$ A system of linear equations is said to be consistent, if it possess the solution.
Otherwise it is inconsistent.

* A matrix form of system of linear equations:
In the above system of linear equations

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Consider a linear homogeneous equations (2)

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0$$

$$a_{12}x_1 + a_{22}x_2 + \dots + a_{n2}x_n = 0$$

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0$$

are of 'm'-homogeneous linear equations in n-unknowns i.e., $x_1, x_2, x_3, \dots, x_n$.

∴ Here the above system can be written as $Ax = 0$

where $O = \text{zero matrix of order } m \times n$.

In the above system is always consistent as $x_1=0, x_2=0, x_3=0, \dots, x_n=0$ is always a solution of equation (2).

These solutions are called trivial solutions.

Any other solution is called non-trivial solution.

v. If $r \neq n$ then the solution will depend on $n-r$ parameters in terms of which the complete solution can be obtained.

Q.1. $x + y - 6z = 0$

$$-3x + y + 2z = 0$$

$$x - y + 2z = 0$$

$$Ax = 0$$

$$\begin{bmatrix} 1 & 1 & -6 \\ -3 & 1 & 2 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R \rightarrow R + 3(R)$$

$$-3 \rightarrow -3 + 3(1) = 0$$

$$1 \rightarrow 1 + 3(1) = 4$$

$$2 \rightarrow 2 + 3(-6) = -16$$

$$R \rightarrow R + 3(R)$$

~~$1^3 - 1 + 3(1) = 4$~~

$$-1 \rightarrow -1 + 3(1) = 2$$

$$2 \rightarrow 2 + 3(-6) = -16$$

$$R \rightarrow R^{-1}(R)$$

3 3

$$\Rightarrow \begin{bmatrix} 1 & 1 & -6 \\ 0 & 4 & -16 \\ 9 & 2 & -16 \end{bmatrix}$$

①, sub y and z

$$x + 4t - 6t = 0$$

$$x - 2t = 0$$

$$\therefore x = 2t //$$

$$20. \quad 3x_1 = 0$$

$$-x_1 + 4x_2 = 0$$

$$4x_1 - 4x_2 + 3x_3 = 0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 4 & -4 & 3 \end{bmatrix} \sim \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 1(R_1)$$

$$-1 \rightarrow -1 + 1(1) = 0$$

$$4 \rightarrow 4 + 1(0) = 4$$

$$0 \rightarrow 0 + 1(0) = 0$$

$$R_3 \rightarrow R_3 - 4(R_2)$$

$$4 \rightarrow 4 - 4(1) = 0$$

$$-4 \rightarrow -4 - 4(0) = -4$$

$$3 \rightarrow 3 - 4(0) = 3$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & -4 & 3 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{4}(R_2)$$

$$0 \rightarrow \frac{1}{4}(0) = 0$$

$$4 \rightarrow \frac{1}{4}(4) = 1$$

$$0 \rightarrow \frac{1}{4}(0) = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 3 \end{bmatrix}$$

Sol:
$$\begin{bmatrix} 1 & -3 & 2 \\ 7 & -21 & 14 \\ -3 & 9 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{array}{l|l} R \rightarrow R - 7(R_1) & R \rightarrow R + 3(R_1) \\ 7 \xrightarrow{2} 7 - 7(1) = 0 & -3 \xrightarrow{3} -3 + 3(1) = 0 \\ -21 \xrightarrow{-3} -21 - 7(-3) = 0 & 9 \xrightarrow{3} 9 + 3(-3) = 0 \\ 14 \xrightarrow{2} 14 - 7(2) = 0 & -6 \xrightarrow{3} -6 + 3(2) = 0 \end{array}$$

$\Rightarrow \begin{bmatrix} 1 & -3 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$x - 3y + 2z = 0$$

Consider $y = t$, and $z = t_2$

$$x - 3t_1 + 2t_2 = 0$$

$$x = 3t_1 - 2t_2 //$$

4. $x_1 + x_2 - x_3 + x_4 = 0$

$$x_1 - x_2 + 2x_3 - x_4 = 0$$

$$3x_1 + x_2 + x_4 = 0$$

Sol:
$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{array}{l|l} R_2 \rightarrow R_2 - 1(R_1) & R_3 \rightarrow R_3 - 3(R_1) \\ 1 \xrightarrow{1} 1 - 1(1) = 0 & 3 \xrightarrow{3} 3 - 3(1) = 0 \\ -1 \xrightarrow{-1} -1 - 1(1) = -2 & 1 \xrightarrow{1} 1 - 3(1) = -2 \\ 2 \xrightarrow{2} 2 - 1(-1) = 3 & 0 \xrightarrow{0} 0 - 3(1) = -3 \\ -1 \xrightarrow{-1} -1 - 1(1) = -2 & 1 \xrightarrow{1} 1 - 3(1) = -2 \end{array}$$

system of non-homogeneous linear equations
consider the system of m non-homogeneous equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

The above system can be written in the matrix form
i.e., $AX=B$ where A = co-efficient matrix,
 X = matrix of unknown
 B = matrix of independent terms

then

The augmented matrix is given by $[A:B] =$

$$[A:B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & : & b_2 \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & : & b_m \end{bmatrix}$$

Working rule to find the solution of non-homogeneous linear equation:

Consider the system of equation $AX=B$

Find the rank of the augmented matrix $A:B$ by echelon form

Then from the above step rank of A and rank of $A:B$ can be obtained

If $\rho(A) < \rho(A:B)$ then the system is inconsistent and therefore it has no solution

If $\rho(A) = \rho(A:B)$ then the system is consistent, we can find the solution by considering the system of equations

$$\begin{aligned} x + y - 2z &= 5 \\ x - 2y + z &= -2 \\ -2x + y + z &= 4 \end{aligned}$$

Sol: Consider $AX=B$

$$\begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 4 \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} 1 & 1 & -2 & : & 5 \\ 1 & -2 & 1 & : & -2 \\ -2 & 1 & 1 & : & 4 \end{bmatrix}$$

$$\begin{array}{ll} R \rightarrow R - 1(R) & R \rightarrow R + 2(R) \\ 1 \rightarrow 1 - 1(1) = 0 & 2 \rightarrow -2 + 2(1) = 0 \\ -2 \rightarrow -2 - 1(1) = -3 & 1 \rightarrow 1 + 2(1) = 3 \\ 1 \rightarrow 1 - 1(-2) = 3 & 1 \rightarrow 1 + 2(-2) = -3 \\ -2 \rightarrow -2 - 1(5) = -7 & 4 \rightarrow 4 + 2(5) = 14 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -2 & : & 5 \\ 0 & -3 & 3 & : & -7 \\ 0 & 3 & -3 & : & 14 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{-3}(R_2)$$

$$0 \rightarrow \frac{1}{-3}(0) = 0$$

$$-3 \rightarrow \frac{1}{-3}(-3) = 1$$

$$3 \rightarrow \frac{1}{-3}(3) = -1$$

$$-7 \rightarrow \frac{1}{-3}(-7) = 2.33$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & -2 & 3 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad (R_2)$$

$$0 \rightarrow 1 \quad (0) = 0$$

$$-2 \rightarrow 1 \quad (-2) = 1$$

$$3 \rightarrow 1 \quad (3) = -1.5$$

$$-2 \rightarrow 1 \quad (-2) = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & -1.5 & 1 \\ 0 & -2 & 3 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2(R_2)$$

$$0 \rightarrow 0 + 2(0) = 0$$

$$-2 \rightarrow -2 + 2(1) = 0$$

$$3 \rightarrow 3 + 2(-1.5) = 0$$

$$-2 \rightarrow -2 + 2(1) = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & -1.5 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$x_1 + x_2 - x_3 + x_4 = 0 \rightarrow (1)$$

$$x_2 - 1.5x_3 + x_4 = 0 \rightarrow (2)$$

$$x_2 = 1.5x_3 - x_4$$

$$\begin{bmatrix} 1 & 1 & -2 & : & 5 \\ 0 & 1 & -1 & : & 2.33 \\ 0 & 3 & -3 & : & 14 \end{bmatrix}$$

$$\begin{aligned} R_3 &\rightarrow R_3 - 3(R_2) \\ 0 &\rightarrow 0 - 3(0) = 0 \\ 3 &\rightarrow 3 - 3(1) = 0 \\ 3 &\rightarrow 3 - 3(-1) = 0 \\ 14 &\rightarrow 14 - 3(2.33) = -20.99 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & -2 & : & 5 \\ 0 & 1 & -1 & : & 2.33 \\ 0 & 0 & 0 & : & -20.99 \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{3} (R_3)$$

$$0 \rightarrow \frac{1}{-20.99} (0) = 0$$

$$0 \rightarrow \frac{1}{-20.99} (0) = 0$$

$$0 \rightarrow \frac{1}{-20.99} (0) = 0$$

$$-20.99 \rightarrow \frac{1}{-20.99} (-20.99) = 1$$

$$\begin{bmatrix} 1 & 1 & -2 & : & 5 \\ 0 & 1 & -1 & : & 2.33 \\ 0 & 0 & 0 & : & 1 \end{bmatrix}$$

$$S(A) = 2$$

$$S(A:B) = 3$$

$$\therefore S(A) < S(A:B)$$

$\therefore$ The system is inconsistent and has no solution

$$\begin{aligned} Q2. \quad x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

$$\text{Sol: Consider } AX = B$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 1 \\ 0 \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 2 & 4 & -3 & : & 1 \\ 3 & 6 & -5 & : & 0 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 2(R_1) & R_3 &\rightarrow R_3 - 3(R_1) \\ 2 &\rightarrow 2 - 2(1) = 0 & 3 &\rightarrow 3 - 3(1) = 0 \\ 4 &\rightarrow 4 - 2(1) = 2 & 6 &\rightarrow 6 - 3(1) = 3 \\ -3 &\rightarrow -3 - 2(2) = -7 & -5 &\rightarrow -5 - 3(2) = -11 \\ 1 &\rightarrow 1 - 2(9) = -17 & 0 &\rightarrow 0 - 3(9) = -27 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & 2 & : & 9 \\ 0 & 2 & -7 & : & -17 \\ 0 & 3 & -11 & : & -27 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{2} (R_2)$$

$$0 \rightarrow \frac{1}{2} (0) = 0$$

$$2 \rightarrow \frac{1}{2} (2) = 1$$

$$-7 \rightarrow \frac{1}{2} (-7) = -3.5$$

$$-17 \rightarrow \frac{1}{2} (-17) = -8.5$$

$$z = 0.07 \leftarrow (3)$$

Substitute (3) in (5)

$$y - 3.5(0.07) = -8.5$$

$$y - 0.245 = -8.5$$

$$y = -8.255 \approx -8.3$$

Now,

Substitute y, z in (1)

$$x - 8.3 + 2(0.07) = 9$$

$$x - 8.3 - 0.14 = 9$$

$$x = 9 + 8.44$$

$$x = 0.56$$

Now,

Sub y, z in (2)

$$7 - 8.3 + 2(0.07) = 9$$

$$7 - 8.3 - 0.14 = 9$$

$$x = 9 + 8.44$$

$$x = 0.56 \approx 1$$

$$\therefore x = 1, y = -8.3, z = 0.07 //$$

* Questions:

1. Find the values of λ and μ such that the following system of equations has

(i) No solution

(ii) Unique solution

(iii) Infinitely many solution

$$x + 3y + 4z = 5;$$

$$x + 2y + z = 3;$$

$$x + 3y + \lambda z = \mu.$$

2. Find whether the following system possess a non-trivial solution

$$x - 3y + 2z = 0$$

$$7x - 21y + 14z = 0$$

$$-3x + 9y - 6z = 0$$

Sol:
$$\begin{bmatrix} 1 & -3 & 2 \\ 7 & -21 & 14 \\ -3 & 9 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

①
8.5 $\rightarrow$ ②

$$\begin{array}{l}
 R_1 \rightarrow R_1 - 7(R_2) \\
 3 \rightarrow 3 - 7(1) = 0 \\
 7 \rightarrow 7 - 7(-3) = 0 \\
 -21 \rightarrow -21 - 7(-3) = 0 \\
 14 \rightarrow 14 - 7(2) = 0
 \end{array}
 \quad
 \begin{array}{l}
 R_3 \rightarrow R_3 + 3(R_2) \\
 -3 \rightarrow -3 + 3(1) = 0 \\
 9 \rightarrow 9 + 3(-3) = 0 \\
 -6 \rightarrow -6 + 3(2) = 0
 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Yes, it possess non-trivial solution //

3. What is consistent linear equation? Solve completely the following system of equations $2x - y + 3z = 0$, $3x + 2y + z = 0$, $x - 4y + 5z = 0$

Sol:
$$\begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & -4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Interchange $R_1 \leftrightarrow R_3$

$$\Rightarrow \begin{bmatrix} 1 & -4 & 5 \\ 3 & 2 & 1 \\ 2 & -1 & 3 \end{bmatrix}$$

$$\begin{array}{l}
 R_1 \rightarrow R_1 - 3(R_2) \\
 3 \rightarrow 3 - 3(1) = 0 \\
 2 \rightarrow 2 - 3(-4) = 14 \\
 1 \rightarrow 1 - 3(5) = -14
 \end{array}
 \quad
 \begin{array}{l}
 R_2 \rightarrow R_2 - 2(R_1) \\
 2 \rightarrow 2 - 2(1) = 0 \\
 -1 \rightarrow -1 - 2(-4) = 7 \\
 3 \rightarrow 3 - 2(5) = -7
 \end{array}$$

$$\begin{bmatrix} 1 & -4 & 5 \\ 0 & 14 & -14 \\ 0 & 7 & -7 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - \frac{1}{2}(R_2)$$

$$14 \rightarrow \frac{1}{14}(14) = 1$$

$$3 \rightarrow \frac{1}{14}(3) = 0.21$$

$$\begin{bmatrix} 1 & -4 & 5 \\ 0 & 1 & 0.21 \\ 0 & 7 & -7 \end{bmatrix}$$

$$\begin{array}{l}
 R_1 \rightarrow R_1 - 7(R_2) \\
 0 \rightarrow 0 - 7(0.21) = -1.47 \\
 7 \rightarrow 7 - 7(1) = 0 \\
 -7 \rightarrow -7 - 7(0.21) = -1.47
 \end{array}$$

$$\begin{bmatrix} 1 & -4 & 5 \\ 0 & 1 & 0.21 \\ 0 & 0 & -1.47 \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{-1.47}(R_3)$$

$$0 \rightarrow \frac{1}{-1.47}(0) = 0$$

$$0 \rightarrow \frac{1}{-1.47}(0) = 0$$

$$-1.47 \rightarrow \frac{1}{-1.47}(-1.47) = 1$$

$$\begin{bmatrix} 1 & -4 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$-7y + 5z = 0 \rightarrow (1)$$

$$y = 0 \rightarrow (2)$$

$$z = 0$$

$\therefore$ H is trivial solution.